Electrostatic Problem of a Charged Disc

1 Statement of the Problem

The potential and capacitance of a thin, charged circular conducting disc (Problem 3.3 in *Classical Electrodynamics* by Jackson [1]) is arguably one of the most difficult problems in the book. For me, the main difficulty arises from the way the problem is described: it states that the charge is distributed over the entire surface of the disc, rather than concentrated on its edge, as one might expect for a perfect conductor. In electrostatic equilibrium, the charge on a conductor resides on its boundary (see Problem 1.1(a)). This apparent contradiction is beautifully explained by Michael Fowler [2], who points out that the migration of charge to the surface of a conductor follows directly from the inverse-square distance dependence of Coulomb's law. Crucially, this argument holds only in three dimensions. Once this issue concerning the charge distribution on the disc is clarified, the problem can be solved straightforwardly using the method described in Section 3.3.

In this note, I present several additional methods for solving this problem. As is common in electrostatic problems with different geometries, I consider two orthogonal coordinate systems: ellipsoidal and oblate spheroidal coordinates. Although the full solution of the Laplace's equation can in principle be obtained by separation of variables in these coordinate systems, I focus only on particular solutions that arise naturally from families of equipotential surfaces.

2 Ellipsoidal Coordinates

A circular disc can be regarded as the limiting case of an oblate spheroid that has been flattened along its symmetry axis. Since an oblate spheroid is a special type of ellipsoid, the disc may therefore be viewed as a degenerate ellipsoid. Consequently, one can solve the electrostatic problem for a general ellipsoid and then obtain the corresponding result for a circular disc by taking an appropriate limiting case.

2.1 Scaled spherical coordinates

An ellipsoid can be described by the equation

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1. \tag{1}$$

By analogy with spherical coordinates, one may parameterize the Cartesian coordinates as

$$x = ar\sin\theta\cos\varphi, \qquad y = br\sin\theta\sin\varphi, \qquad z = cr\cos\theta,$$

where $r = 1$, $\theta \in [0, \pi]$, and $\varphi \in [0, 2\pi]$. However, the parameters r, θ, and φ do *not* form an orthogonal coordinate system. The parametrization can be viewed as the mapping

$$\mathbf{r}(r, \theta, \varphi) = \big(ar\sin\theta\cos\varphi,\ br\sin\theta\sin\varphi,\ cr\cos\theta\big).$$

The corresponding coordinate basis vectors are

$$\mathbf{r}_r = (a\sin\theta\cos\varphi,\ b\sin\theta\sin\varphi,\ c\cos\theta),$$

$$\mathbf{r}_\theta = (ar\cos\theta\cos\varphi,\ br\cos\theta\sin\varphi,\ -cr\sin\theta),$$

$$\mathbf{r}_\varphi = (-ar\sin\theta\sin\varphi,\ br\sin\theta\cos\varphi,\ 0).$$

For an orthogonal coordinate system, these vectors must be mutually perpendicular. However, their dot products are generally nonzero. For example,

$$\mathbf{r}_r \cdot \mathbf{r}_\theta = r\sin\theta\cos\theta\left(a^2\cos^2\varphi + b^2\sin^2\varphi - c^2\right),$$

which does not vanish in general. Similar calculations show that

$$\mathbf{r}_r \cdot \mathbf{r}_\varphi \neq 0, \qquad \mathbf{r}_\theta \cdot \mathbf{r}_\varphi \neq 0.$$

Therefore, the coordinates (r, θ, φ) are not orthogonal. Only in the special case $a = b = c$, corresponding to a sphere, do these expressions reduce to the standard spherical coordinates, which are orthogonal. Geometrically, the parametrization arises from scaling spherical coordinates by different factors along the x, y, and z directions. Such anisotropic scaling does not preserve angles, so the originally orthogonal spherical grid becomes skewed on the ellipsoid. Non-orthogonal coordinate systems complicate the expression of the Laplace operator. Consequently, the scaled spherical coordinates introduced above are not well suited for solving this problem.

2.2 Ellipsoidal coordinates

Ellipsoidal coordinates form an orthogonal curvilinear coordinate system naturally adapted to problems involving ellipsoids and related quadratic surfaces. They are particularly useful in solving the Laplace equation for boundary-value problems where the boundary is an ellipsoid. Ellipsoidal coordinates (ξ, η, ζ) are defined as the three roots u of

$$\frac{x^2}{a^2+u} + \frac{y^2}{b^2+u} + \frac{z^2}{c^2+u} = 1,$$

where $a > b > c > 0$. For any point (x, y, z), the three solutions satisfy

$$-a^2 \leq \zeta \leq -b^2 \leq \eta \leq -c^2 \leq \xi.$$

The ellipsoidal coordinates have a clear geometric interpretation. Surfaces of constant ξ satisfy

$$\frac{x^2}{a^2+\xi} + \frac{y^2}{b^2+\xi} + \frac{z^2}{c^2+\xi} = 1$$

represent a family of ellipsoids. Similarly, surfaces of constant η

$$\frac{x^2}{a^2+\eta} + \frac{y^2}{b^2+\eta} + \frac{z^2}{c^2+\eta} = 1$$

correspond to one-sheeted hyperboloids, and surfaces of constant ζ

$$\frac{x^2}{a^2+\zeta} + \frac{y^2}{b^2+\zeta} + \frac{z^2}{c^2+\zeta} = 1$$

represent two-sheeted hyperboloids. All these families of quadric surfaces are confocal, sharing the same focal structure as the ellipsoid given in (1). A point in three-dimensional space can be uniquely specified by the intersection of these three surfaces.

The transformation from ellipsoidal coordinates (ξ, η, ζ) to Cartesian coordinates is

$$x^2 = \frac{(a^2+\xi)(a^2+\eta)(a^2+\zeta)}{(a^2-b^2)(a^2-c^2)}, \qquad y^2 = \frac{(b^2+\xi)(b^2+\eta)(b^2+\zeta)}{(b^2-a^2)(b^2-c^2)}, \qquad z^2 = \frac{(c^2+\xi)(c^2+\eta)(c^2+\zeta)}{(c^2-a^2)(c^2-b^2)}.$$

For convenience, we introduce the notation

$$R_\xi = (a^2+\xi)(b^2+\xi)(c^2+\xi),$$

and define R_η and R_ζ analogously. In terms of these quantities, the scale factors of the ellipsoidal coordinate system can be written as

$$h_\xi = \frac{1}{2}\sqrt{\frac{(\xi-\eta)(\xi-\zeta)}{R_\xi}}, \quad h_\eta = \frac{1}{2}\sqrt{\frac{(\eta-\xi)(\eta-\zeta)}{R_\eta}}, \quad h_\zeta = \frac{1}{2}\sqrt{\frac{(\zeta-\xi)(\zeta-\eta)}{R_\zeta}}.$$

Using these scale factors, the Laplace operator in ellipsoidal coordinates takes the form

$$\nabla^2 = \frac{1}{h_\xi h_\eta h_\zeta}\left[\frac{\partial}{\partial\xi}\left(\frac{h_\eta h_\zeta}{h_\xi}\frac{\partial}{\partial\xi}\right) + \frac{\partial}{\partial\eta}\left(\frac{h_\xi h_\zeta}{h_\eta}\frac{\partial}{\partial\eta}\right) + \frac{\partial}{\partial\zeta}\left(\frac{h_\xi h_\eta}{h_\zeta}\frac{\partial}{\partial\zeta}\right)\right].$$

2.3 Solving electrostatic problem in ellipsoidal coordinates

In ellipsoidal coordinates, the Laplace equation takes the form

$$\nabla^2\Phi(\xi,\eta,\zeta) = 0,$$

or equivalently,

$$\frac{1}{h_\xi h_\eta h_\zeta}\left[\frac{\partial}{\partial\xi}\left(\frac{h_\eta h_\zeta}{h_\xi}\frac{\partial\Phi}{\partial\xi}\right) + \frac{\partial}{\partial\eta}\left(\frac{h_\xi h_\zeta}{h_\eta}\frac{\partial\Phi}{\partial\eta}\right) + \frac{\partial}{\partial\zeta}\left(\frac{h_\xi h_\eta}{h_\zeta}\frac{\partial\Phi}{\partial\zeta}\right)\right] = 0.$$

This equation can be solved by the method of separation of variables. The resulting solutions are known as ellipsoidal harmonics, which play a role analogous to spherical harmonics in problems with ellipsoidal symmetry.

However, for an equipotential ellipsoid with $\xi = 0$, we an assume that the solution only depends on ξ, *i.e.*, $\Phi(\xi,\eta,\zeta) \equiv \Phi(\xi)$. Under this assumption, the Laplace equation simplifies to

$$\frac{\mathrm{d}}{\mathrm{d}\xi}\left(\sqrt{R_\xi}\frac{\mathrm{d}\Phi}{\mathrm{d}\xi}\right) = 0.$$

The general solution of this equation is

$$\Phi(\xi) = A\int_\xi^{+\infty}\frac{\mathrm{d}\xi'}{\sqrt{R_{\xi'}}} = A\int_\xi^{+\infty}\frac{\mathrm{d}\xi'}{\sqrt{(a^2+\xi')(b^2+\xi')(c^2+\xi')}},$$

where the constant A is determined by boundary conditions. This solution naturally satisfies $\Phi \to 0$ as $\xi \to +\infty$.

For an ellipsoid carrying a total charge Q, the potential at large distances behaves like

$$\Phi \sim \frac{Q}{4\pi\varepsilon_0 r}.$$

In ellipsoidal coordinates, $r \to +\infty$ corresponds to $\xi \to +\infty$, since $r \sim \xi^2$. At large ξ, the factor under the integral behaves as

$$R_{\xi'} \sim (\xi')^3, \quad \text{so that} \quad \sqrt{R_{\xi'}} \sim (\xi')^{3/2}.$$

Therefore, for large ξ, the potential behaves as

$$\Phi(\xi) \sim \frac{2A}{\sqrt{\xi}} \sim \frac{2A}{r}.$$

Comparing with the known asymptotic form, we find

$$A = \frac{Q}{8\pi\varepsilon_0},$$

and the potential of the charged ellipsoid can therefore be written as

$$\Phi(\xi) = \frac{Q}{8\pi\varepsilon_0} \int_{\xi}^{+\infty} \frac{\mathrm{d}\xi'}{\sqrt{R_{\xi'}}}.$$

On the ellipsoidal surface, where $\xi = 0$, the potential becomes

$$\Phi(0) = \frac{Q}{8\pi\varepsilon_0} \int_{0}^{+\infty} \frac{\mathrm{d}\xi}{\sqrt{R_{\xi}}}.$$

From this, the capacitance of the ellipsoid is given by

$$\frac{1}{C} = \frac{1}{8\pi\varepsilon_0} \int_{0}^{+\infty} \frac{\mathrm{d}\xi}{\sqrt{R_{\xi}}}.$$

The surface charge density can be determined from the normal derivative of the potential

$$\sigma = -\varepsilon_0 \left.\frac{\partial\Phi}{\partial\xi}\right|_{\xi=0} = -\varepsilon_0 \left(\frac{1}{h_\xi}\frac{\partial\Phi}{\partial\xi}\right)_{\xi=0} = \frac{Q}{4\pi\sqrt{\eta\zeta}}.$$

Notice that on the surface $\xi = 0$, we have

$$\frac{x^2}{a^4} + \frac{y^2}{b^4} + \frac{z^2}{c^4} = \frac{\eta\zeta}{a^2b^2c^2}.$$

Therefore, the surface charge density can be expressed as

$$\sigma = \frac{Q}{4\pi abc}\left(\frac{x^2}{a^4} + \frac{y^2}{b^4} + \frac{z^2}{c^4}\right)^{-1/2}. \tag{2}$$

For an oblate spheroid, where $a = b > c$, the potential and capacitance can be expressed in closed form. The potential is

$$\Phi(\xi) = \frac{Q}{8\pi\varepsilon_0} \int_{\xi}^{+\infty} \frac{\mathrm{d}\xi'}{\sqrt{R_{\xi'}}} = \frac{Q}{8\pi\varepsilon_0} \int_{\xi}^{+\infty} \frac{\mathrm{d}\xi'}{(a^2+\xi')\sqrt{c^2+\xi'}}$$

$$
\begin{aligned}
&= \frac{Q}{4\pi\varepsilon_0}\int_{\xi}^{+\infty}\frac{\mathrm{d}(\sqrt{c^2+\xi'})}{a^2-c^2+c^2+\xi'} \\
&= \frac{Q}{4\pi\varepsilon_0\sqrt{a^2-c^2}}\arctan\sqrt{\frac{\xi'+c^2}{a^2-c^2}}\Bigg|_{\xi}^{+\infty} \\
&= \frac{Q}{4\pi\varepsilon_0\sqrt{a^2-c^2}}\left(\frac{\pi}{2}-\arctan\sqrt{\frac{\xi+c^2}{a^2-c^2}}\right) \\
&= \frac{Q}{4\pi\varepsilon_0\sqrt{a^2-c^2}}\arctan\sqrt{\frac{a^2-c^2}{\xi+c^2}},
\end{aligned}
$$

where we have used the relation

$$\arctan x + \arctan\left(\frac{1}{x}\right) = \frac{\pi}{2}.$$

The capacitance is then

$$C = \frac{Q}{\Phi(0)} = \frac{4\pi\varepsilon_0\sqrt{a^2-c^2}}{\arctan\sqrt{\frac{a^2}{c^2}-1}} = \frac{4\pi\varepsilon_0\sqrt{a^2-c^2}}{\arccos(c/a)}.$$

In the special case of a circular disc, where $a = b$ and $c = 0$, the capacitance reduces to

$$C = 8\varepsilon_0 a.$$

Taking the limit in Eq. (2) as $c \to 0$ and $z \to 0$, while maintaining the ratio as

$$\frac{z}{c} = \sqrt{1-\frac{\rho^2}{a^2}}, \qquad \rho^2 = x^2 + y^2,$$

the surface charge density becomes

$$\sigma = \frac{Q}{4\pi a^2}\left(1-\frac{\rho^2}{a^2}\right)^{-1/2}.$$

Notice that this is the charge density on one side of the disc, and total charge density of the disc is twice of this value.

3 Oblate Spheroidal Coordinates

In the previous section, I treated a circular disc as a limiting case of an ellipsoid–more precisely, an oblate spheroid. In this section, I instead begin directly with oblate spheroidal coordinates to solve Laplace equation in this coordinate system. This approach leads naturally to the same result for the circular disc.

3.1 Oblate spheroidal coordinates

Oblate spheroidal coordinates (μ, ν, ϕ) are related to Cartesian coordinates (x, y, z) by

$$x = a\cosh\mu\cos\nu\cos\phi, \qquad y = a\cosh\mu\cos\nu\sin\phi, \qquad z = a\sinh\mu\sin\nu,$$

where $a > 0$ is a constant that sets the focal length or the distance scale, with coordinate ranges $\mu \in [0, +\infty)$, $\nu \in [0, \pi]$, and $\phi \in [0, 2\pi]$. It is readily verified that the surface $\mu = 0$ corresponds to a flat circular disc of radius a lying in the plane $z = 0$.

The scale factors are given by

$$h_\mu = h_\nu = a\sqrt{\sinh^2 \mu + \sin^2 \nu}, \qquad h_\phi = a \cosh \mu \cos \nu.$$

The Laplace operator in the oblate spheroidal coordinates is

$$\nabla^2 = \frac{1}{a^2(\sinh^2 \mu + \sin^2 \nu)} \left[\frac{1}{\cosh \mu} \frac{\partial}{\partial \mu} \left(\cosh \mu \frac{\partial}{\partial \mu}\right) + \frac{1}{\cos \nu} \frac{\partial}{\partial \nu} \left(\cos \nu \frac{\partial}{\partial \nu}\right)\right] + \frac{1}{a^2 \cosh^2 \mu \cos^2 \nu} \frac{\partial^2}{\partial \phi^2}.$$

3.2 Solving electrostatic problem in oblate spheroidal coordinates

In oblate spheroidal coordinates, the Laplace equation takes the form

$$\nabla^2 \Phi(\mu, \nu, \phi) = 0,$$

or equivalently,

$$\frac{1}{a^2(\sinh^2 \mu + \sin^2 \nu)} \left[\frac{1}{\cosh \mu} \frac{\partial}{\partial \mu} \left(\cosh \mu \frac{\partial \Phi}{\partial \mu}\right) + \frac{1}{\cos \nu} \frac{\partial}{\partial \nu} \left(\cos \nu \frac{\partial \Phi}{\partial \nu}\right)\right] + \frac{1}{a^2 \cosh^2 \mu \cos^2 \nu} \frac{\partial^2 \Phi}{\partial \phi^2} = 0.$$

Rather than solving the full equation, we restrict attention to solutions that depend only on μ, *i.e.*, $\Phi(\mu, \nu, \phi) \equiv \Phi(\mu)$. Under this assumption, the Laplace equation reduces to

$$\frac{\partial}{\partial \mu} \left(\cosh \mu \frac{\partial \Phi}{\partial \mu}\right) = 0,$$

whose general solution is

$$\Phi(\mu) = A \int_\mu^{+\infty} \frac{\mathrm{d}\mu'}{\cosh \mu'} = 2A \arctan e^{\mu'} \Big|_\mu^{+\infty} = A(\pi - 2 \arctan e^\mu),$$

and the upper limit has been chosen so that the potential vanishes as $\mu \to +\infty$.

To determine the constant A, we evaluate the potential at $\mu = 0$,

$$V = \Phi(0) = A(\pi - 2 \arctan 1) = \frac{\pi}{2} A,$$

which immediately gives

$$A = \frac{2V}{\pi}.$$

Hence,

$$\Phi(\mu) = V \left(2 - \frac{4}{\pi} \arctan e^\mu\right).$$

To obtain a more symmetric form, we use the identity

$$\arctan e^\mu + \arctan e^{-\mu} = \frac{\pi}{2},$$

and

$$\tan(\arctan e^\mu - \arctan e^{-\mu}) = \frac{e^\mu - e^{-\mu}}{2} = \sinh \mu,$$

to express the potential in a compact form,

$$\Phi(\mu) = V\left(1 - \frac{2}{\pi}\arctan\sinh\mu\right).$$

The surface charge density on the disc is obtained from the normal derivative of the potential

$$\sigma = -\varepsilon_0\left(\frac{1}{h_\mu}\frac{\partial\Phi}{\partial\mu}\right)_{\mu=0} = \frac{2\varepsilon_0 V}{\pi a\sin\nu},$$

and the total charge on the disc is

$$Q = \int \sigma dS = \int_0^\pi \mathrm{d}\nu \int_0^{2\pi} \mathrm{d}\phi\ \sigma h_\nu h_\phi = 8\varepsilon_0 aV.$$

Finally, the capacitance is then

$$C = \frac{Q}{V} = 8\varepsilon_0 a.$$

The charge density on the disc can also be expressed in cylindrical coordinates. On the surface of the disc, $\rho = a\cos\nu$, so that

$$\sin\nu = \sqrt{1 - \frac{\rho^2}{a^2}}.$$

Substituting into the expression for the charge density, we obtain

$$\sigma(\rho) = \frac{2\varepsilon_0 V}{\pi a}\left(1 - \frac{\rho^2}{a^2}\right)^{-1/2}.$$

Using $Q = 8\varepsilon_0 aV$, this can be written equivalently as

$$\sigma(\rho) = \frac{Q}{4\pi a^2}\left(1 - \frac{\rho^2}{a^2}\right)^{-1/2}.$$

which agrees with the result obtained in the previous section.

References

[1] J.D. Jackson, *Classical Electrodynamics*, 3rd ed. (John Wiley & Sons, 1999).

[2] Michael Fowler, Field of a Charged Conducting Disc.

[3] L. Landau and E. Lifshitz, *Electrodynamics of Continuous Media*, 2nd revised ed. (Butterworth-Heinemann Ltd., 1984).